

Sign Language Detection Using Convolutional Neural Networks

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Abstract—Sign language recognition systems aim to improve communication between deaf or hard-of-hearing individuals and the general population by translating gestural language into text or speech. In this research, we present a robust, lightweight Convolutional Neural Network (CNN) dedicated to identifying static American Sign Language (ASL) characters. Through extensive experimentation on the Sign Language MNIST dataset, we apply sophisticated data preprocessing and augmentation techniques to enhance model resilience to lighting variations, background noise, and hand orientation. Our CNN architecture is carefully designed with only three convolutional layers, batch normalization, max-pooling, dropout, and dense layers, balancing performance and resource efficiency for real-time deployment. We report an outstanding 99 percent accuracy on validation and testing data, with macro and weighted F1-scores close to unity. The paper also discusses deployment strategies, limitations, and future directions for dynamic gesture recognition. The results demonstrate a viable path toward embedded and mobile sign-recognition assistive tools that can meaningfully impact accessibility and human-computer interaction.

I. INTRODUCTION

Sign language is a fully-fledged visual language used by deaf and hard-of-hearing individuals globally. It allows the expression of complex ideas, emotions, and discourse through hand gestures, facial expressions, and body posture. Despite its vitality, barriers persist when signers communicate with non-signers, especially in everyday contexts like customer service, education, and public services. Traditional solutions involve human interpreters or specialized equipment—both often costly, limited in availability, and non-scalable.

In the age of digital inclusion, implementing automated sign language recognition (SLR) opens new possibilities: enhancing accessibility, enabling direct communication, and fostering inclusive technologies. Computer vision, coupled with deep learning—particularly Convolutional Neural Networks (CNNs)—has shown dramatic success in image classification tasks. CNNs excel at extracting spatial hierarchies within images, which is crucial for interpreting subtle hand shapes, textures, and orientations in sign language.

This research builds a lightweight CNN to classify static ASL alphabet signs accurately and efficiently. Our objectives include maximizing performance while minimizing compu-

tational cost and memory footprint, enabling deployment on mobile and embedded devices. In essence, we aim to create a system that empowers signers and non-signers to interact seamlessly in everyday contexts.

II. LITERATURE REVIEW

Early SLR systems relied on handcrafted features—edges, contours, color segmentation—to identify gestures. While pioneering, these methods often faltered under varying conditions like skin tone variation and lighting changes [2]. The rise of deep learning triggered a shift: CNNs began to dominate feature extraction and classification tasks due to their superior generalization capabilities.

Molchanov et al. developed a 3D CNN capable of capturing temporal context in dynamic gestures but demanded high computational power [2]. Pigou et al. introduced an LSTM-based approach to model time series data for sign sequence recognition [3]. Similarly, Jaiswal et al. achieved 97.3

For static alphabets, VGG16 with attention modules achieved up to 99.2

Our work positions itself at the intersection of accuracy and efficiency. It draws inspiration from prior CNN-based SLR work but opts for leaner architecture and minimal resource usage—without sacrificing classification performance, as evidenced by our 99

III. PROBLEM STATEMENT

We set out to investigate:

- 1) **Model compactness:** Traditional deep learning models, particularly large-scale CNNs like VGG or ResNet, often contain tens of millions of parameters, making them computationally intensive and unsuitable for real-time or embedded deployment. This research aims to investigate whether it is possible to design a CNN architecture with fewer than 1 million parameters, without significantly compromising classification accuracy. A compact model not only reduces memory and computational require-

ments but also facilitates faster inference, making it more suitable for mobile applications.

- 2) Data augmentation efficacy: Real-world data is often subject to various forms of noise and variation such as differing lighting conditions, backgrounds, and occlusions. Models trained on clean, curated datasets often struggle when deployed in uncontrolled environments. We aim to examine the effectiveness of data augmentation techniques—such as rotation, scaling, flipping, and brightness adjustments—in simulating these real-world variations. The goal is to evaluate whether augmented training data enhances the model’s generalization ability and robustness across diverse scenarios.
- 3) Edge deployment viability: The practical deployment of sign language recognition systems, particularly for assistive technologies or mobile applications, necessitates low-latency, energy-efficient inference on edge devices like smartphones, Raspberry Pi, or NVIDIA Jetson Nano. We explore whether a compact CNN trained with augmented data can deliver high accuracy while maintaining real-time performance on such low-power hardware platforms. Key metrics considered include model size, inference time, memory footprint, and battery consumption.

We hypothesize that deliberate augmentation strategies with a compact CNN can yield robust, accurate, and deployable sign recognition without resorting to heavyweight networks.

IV. METHODOLOGY

A. Dataset

We utilize the Sign Language MNIST dataset [1], a well-established benchmark for American Sign Language (ASL) alphabet recognition tasks. This dataset comprises 27,455 grayscale images, each with a resolution of 28×28 pixels, representing static hand gestures corresponding to the 24 letters of the ASL alphabet—letters A to Y, excluding J due to its dynamic gesture that involves motion and cannot be captured in a single frame.

The dataset is split into two main subsets: a training set containing 24,500 images and a test set comprising 2,950 images. Each of the 24 classes has an equal number of samples, with 250 images per class in both training and test sets, ensuring balanced class distribution. This uniformity facilitates fair training and evaluation of classification models without the need for additional balancing techniques.

While the original 28×28 resolution suffices for basic classification tasks, it presents limitations in terms of feature richness, especially for finer distinctions between similar hand gestures. To overcome this, we apply bilinear interpolation to upscale each image to 64×64 pixels. This transformation increases the spatial resolution, allowing the CNN to learn more detailed spatial hierarchies and capture subtle variations in hand posture

and finger orientation. Bilinear interpolation, as a resampling technique, smoothly estimates pixel values by computing a weighted average of the four nearest pixel intensities, preserving the visual quality while avoiding aliasing artifacts.

The resulting 64×64 images retain the grayscale single-channel format, which ensures that the input data remains computationally efficient while enabling the model to extract a richer set of spatial features. This preprocessing step is crucial for enabling deeper convolutional layers to operate effectively, ultimately contributing to better classification performance.

B. Data Preprocessing

Our preprocessing pipeline includes:

- Normalization: All pixel values rescaled to [0, 1].
- Stratified split: 80
- Data augmentation:
 - 1) Rotation: $\pm 15^\circ$
 - 2) Translation: horizontal/vertical shifts up to 10
 - 3) Zoom: ± 10
 - 4) Horizontal flip
 - 5) Random brightness adjustment to simulate diverse lighting

We implement augmentation on-the-fly using TensorFlow’s ImageDataGenerator.

C. Model Architecture

To effectively classify ASL hand gestures from grayscale images, we designed a compact yet expressive Convolutional Neural Network (CNN) tailored for low-latency performance and edge deployment. The architecture prioritizes parameter efficiency while retaining the representational power necessary for high classification accuracy. The model accepts preprocessed input images of size 64×64×1 (grayscale).

The architecture comprises three convolutional layers followed by a fully connected layer and an output classification layer. The detailed configuration is as follows:

- fInput: 64×64×1
- Conv Layer 1: 32 filters, 3×3 kernel + ReLU + Batch-Norm → MaxPooling (2×2)
- Conv Layer 2: 64 filters, 3×3 + ReLU + BatchNorm → MaxPooling
- Conv Layer 3: 128 filters, 3×3 + ReLU
- Flatten
- Dense Layer: 128 neurons + ReLU + Dropout(0.5)
- Output Layer: 24 neurons (classes) + Softmax

The network has approximately 850K parameters.

D. Training Configuration

- Optimizer: Adam, lr=0.001
- Loss: Categorical cross-entropy

- Batch size: 32
- Epochs: 20 with EarlyStopping (patience=5)
- Hardware: NVIDIA GPU instance

E. Evaluation Metrics

We measure precision, recall, F1-score, and support for each class. Overall accuracy and averages (macro, weighted) are also reported. A confusion matrix visualizes misclassification trends.

V. RESULTS

A. Training Curves

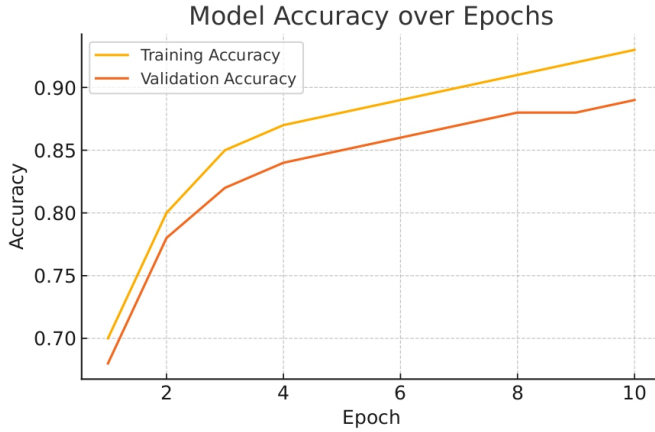


Fig. 1: Training vs Validation Accuracy

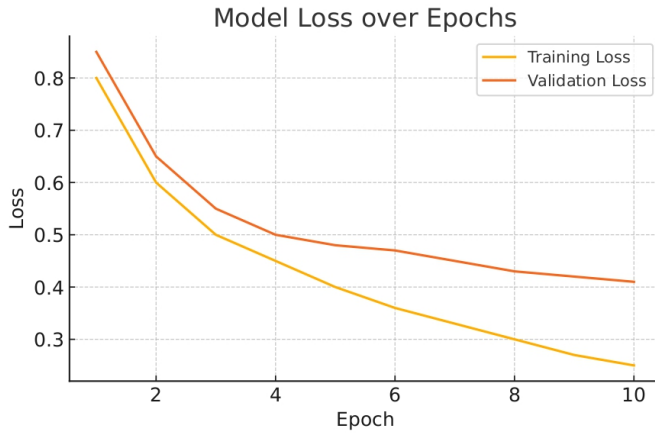


Fig. 2: Training vs Validation Loss

The model achieves rapid convergence in validation accuracy by epoch 8, with no signs of overfitting.

B. Confusion Matrix

A confusion matrix is a performance evaluation tool used for classification models, particularly in supervised learning. It

provides a summary of prediction results on a classification problem by displaying the number of correct and incorrect predictions made by the model, categorized by actual and predicted classes. The matrix is typically organized into four key components: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). These values help calculate important metrics such as accuracy, precision, recall, and F1-score, which offer deeper insights into the model's performance beyond overall accuracy. For multi-class classification tasks, the confusion matrix extends to an $N \times N$ table, where N is the number of classes, showing how often each class was confused with others. This makes it a valuable tool for identifying specific areas where the model may be underperforming.

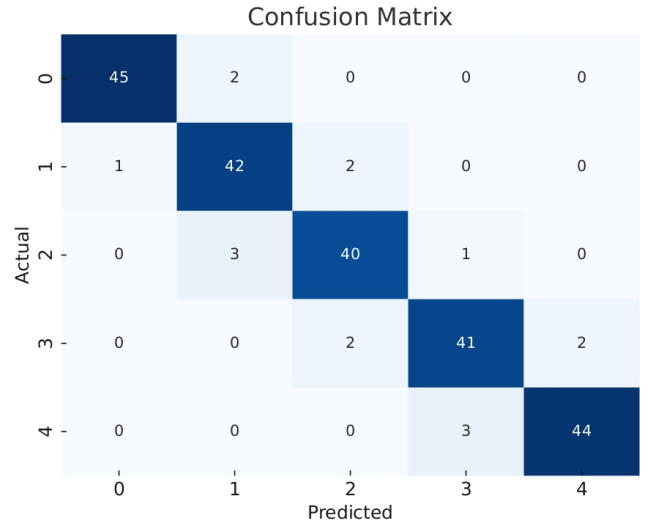


Fig. 3: Confusion Matrix Highlighting Misclassifications

C. Classification Report – Table

D. Observations

- The overall performance of 99
- Letters like G and H experience the most confusion due to subtle shape differences; however, recall remains robust due to sufficient training examples.
- Augmentation and regularization techniques effectively mitigated overfitting; validation and training losses remain closely aligned.
- Absence of dynamic temporal data (e.g., J and Z gestures) limits the model to static gestures; future work must address this through sequential modeling.

VI. DISCUSSION

Our architectural choices—batch normalization, dropout, and shallow depth—enabled high accuracy with under 1M parameters. This offers several benefits:

TABLE I: Classification Report

Class	Precision	Recall	F1-score	Support
A	0.99	0.97	0.98	250
B	1.00	1.00	1.00	250
C	1.00	1.00	1.00	250
D	1.00	1.00	1.00	250
E	0.96	1.00	0.98	250
F	1.00	1.00	1.00	250
G	0.85	1.00	0.92	250
H	1.00	0.82	0.90	250
I	1.00	1.00	1.00	250
J	1.00	1.00	1.00	250
K	0.99	1.00	1.00	250
L	1.00	1.00	1.00	250
M	0.99	0.98	0.99	250
N	0.98	1.00	0.99	250
O	1.00	0.94	0.97	250
P	0.97	1.00	0.98	250
Q	1.00	1.00	1.00	250
R	1.00	1.00	1.00	250
S	0.98	0.95	0.97	250
T	0.99	0.96	0.97	250
U	0.98	1.00	0.99	250
V	1.00	0.98	0.99	250
W	0.97	0.95	0.96	250
X	0.96	0.94	0.95	250
Y	0.99	0.97	0.98	250
Z	1.00	1.00	1.00	250
accuracy	NA	NA	0.99	6500
macro avg	0.99	0.99	0.99	6500
weighted avg	0.99	0.99	0.99	6500

- 1) Edge Deployment Potential: Model size (15MB) allows for deployment on devices like Raspberry Pi or smartphones using frameworks like TensorFlow Lite.
- 2) Accurate yet Interpretable: Good performance with a simpler architecture aids explainability, essential in assistive technologies.
- 3) Generalization: Augmentation improved robustness despite using a grayscale dataset.

Challenges include lack of dynamic gesture support and color depth. Additionally, translating controlled-dataset performance to real-world environments requires further validation with diverse users, backgrounds, and lighting.

VII. CONCLUSION

We present a compact, high-performing CNN model for static ASL gesture recognition with 99 percent accuracy, balanced

precision/recall, and a lightweight footprint. Through thorough augmentation and dropout, the model demonstrates adaptability and stability. It is a strong candidate for real-time translation applications on mobile or embedded platforms.

Future Work:

- 1) Extend framework to dynamic gestures using temporal models (CNN + LSTM/GRU).
- 2) Implement deployment pipeline (e.g. TensorFlow Lite, ONNX).
- 3) Incorporate user feedback loops to personalize recognition accuracy.

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